



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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OUTLINE

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- Methodology
- Results
- Conclusion
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EXECUTIVE SUMMARY

- In this capstone project, we will predict using machine learning if Space X falcon 9 first stage will land successfully
- Methodology followed includes data collection from public Space X API and Space X Wikipedia page. Performed Data wrangling and then exploratory data analysis with data visualization and SQL. Built map with Folium and interactive Dashboard with Plotly Dash. After that predictive analysis was performed using machine learning algorithms.
- Total of four machine learning models were produced that includes Logistic Regression, Support Vector Machine, Decision Tree Classifier, and K Nearest Neighbors. 83.33% accuracy was achieved from two models whereas Decision tree model was found to be most effective predictor.

INTRODUCTION

- **Background:**

The commercial space industry has entered a competitive phase led by SpaceX, which offers rocket launches at around 62 million dollars compared to over 160 million from other providers. The cost advantage comes from its ability to recover and reuse the Falcon 9 first stage. This project focuses on predicting the success of the Falcon 9 first-stage landing using machine learning models. Accurate prediction helps estimate launch costs and supports competitors aiming to match SpaceX's efficiency and pricing.

- **Problem:**

- Which factors such as payload mass, orbit type, and launch site affect the success of the first-stage landing?
- How has the success rate of landings changed over time?
- Which machine learning algorithm provides the highest accuracy in predicting successful landings?
- How can these predictions guide competitive pricing and cost estimation in commercial space launches?

Section 1

Methodology

Methodology

Executive Summary:

This project uses an end-to-end approach to predict the successful landing of the Falcon 9 first stage through data collection, processing, analysis, visualization, and predictive modeling.

- **Data collection methodology:** Data was obtained from the SpaceX API and SpaceX Wikipedia page , containing detailed information on Falcon 9 launches such as dates, launch sites, payloads, and outcomes.
- **Perform data wrangling:** Data cleaning included filling missing values, standardizing formats, and maintaining consistency. Important features were extracted, and new ones were created to enhance the dataset.
- **Perform exploratory data analysis (EDA) using visualization and SQL:**
 - Created visualizations of launch success rates, payloads, and launch sites using Matplotlib and Seaborn.
 - Used SQL queries to extract insights and answer key questions from the dataset.

Methodology

- **Perform interactive visual analytics using Folium and Plotly Dash**
 - Built interactive maps with Folium to show launch sites and outcomes.
 - Developed a Plotly Dash app with dropdowns and sliders to explore launch success rates and payload ranges.
- **Perform predictive analysis using classification models**
 - Trained and tested classification models such as Logistic Regression, SVM, KNN, and Decision Trees.
 - Used GridSearchCV for hyperparameter tuning.
 - Compared model accuracies to identify the best performer for predicting landing success.

Data Collection

- Data was gathered through API requests from the SpaceX REST API and web scraping from the SpaceX Wikipedia page to ensure complete launch information.
- From the SpaceX API, data columns included FlightNumber, Date, BoosterVersion, PayloadMass, Orbit, LaunchSite, Outcome, Flights, GridFins, Reused, Legs, LandingPad, Block, ReusedCount, Serial, Longitude, and Latitude.
- From Wikipedia, extracted fields included Flight Number, Launch site, Payload, PayloadMass, Orbit, Customer, Launch outcome, Version Booster, Booster landing, Date, and Time.

Data Collection – SpaceX API

Step 1: Initiate API Request

Use Python's requests library to connect to the SpaceX API.

Endpoint: <https://api.spacexdata.com/v4/launches>

Step 2: Parse API Response

Convert the JSON response to a Python dictionary.

Extract key fields such as launch date, site, payload mass, rocket type, and outcome.

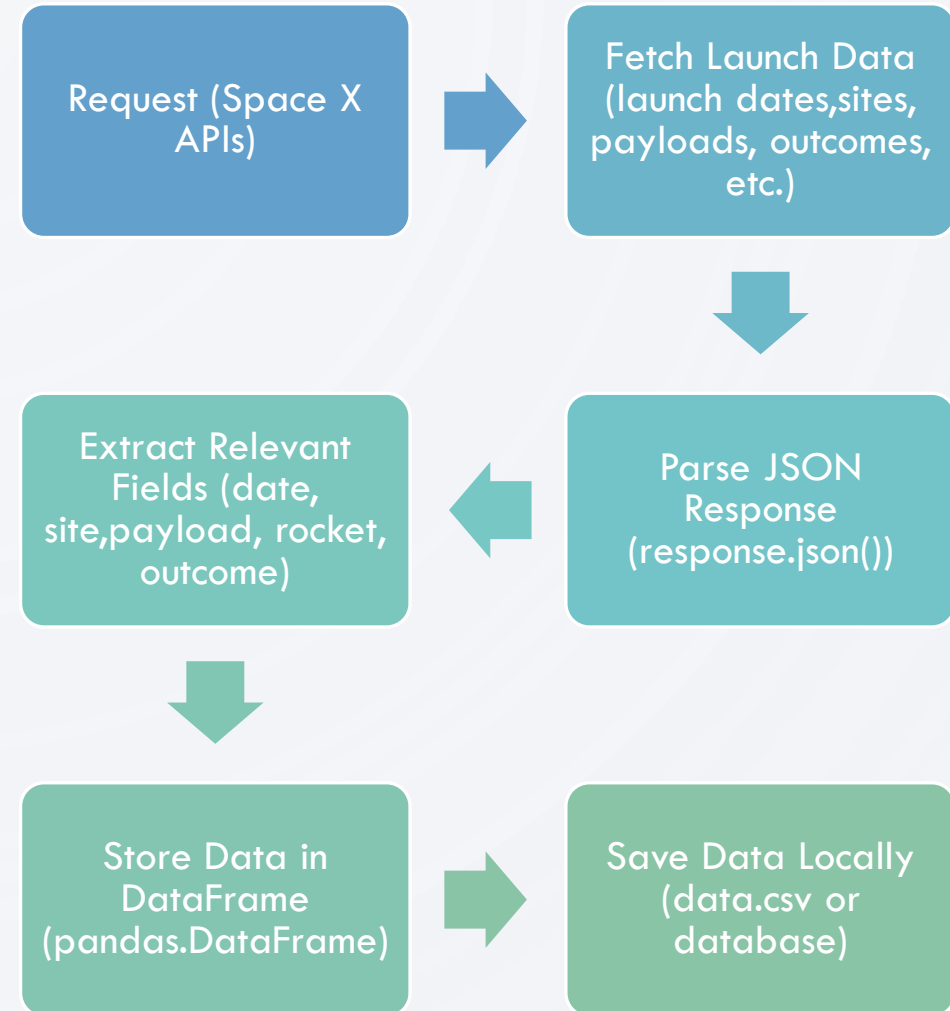
Step 3: Store Data Locally

Save the extracted data into a pandas DataFrame.

Store the DataFrame locally for later processing and analysis.

GitHub URL:

https://github.com/usamacaprian/IBM_Data_Science_Capstone/blob/main/jupyter-labs-spacex-data-collection-api.ipynb



Data Collection - Scraping

Step 1: Initiate Web Scraping

Use Python's `requests` library to fetch the HTML content of the Wikipedia page.

Target URL:

`https://en.wikipedia.org/wiki/List\_of\_Falcon\_9\_and\_Falcon\_Heavy\_launches`

Step 2: Parse HTML Content

Use `BeautifulSoup` to parse the HTML content.

Extract the HTML table containing Falcon 9 launch records.

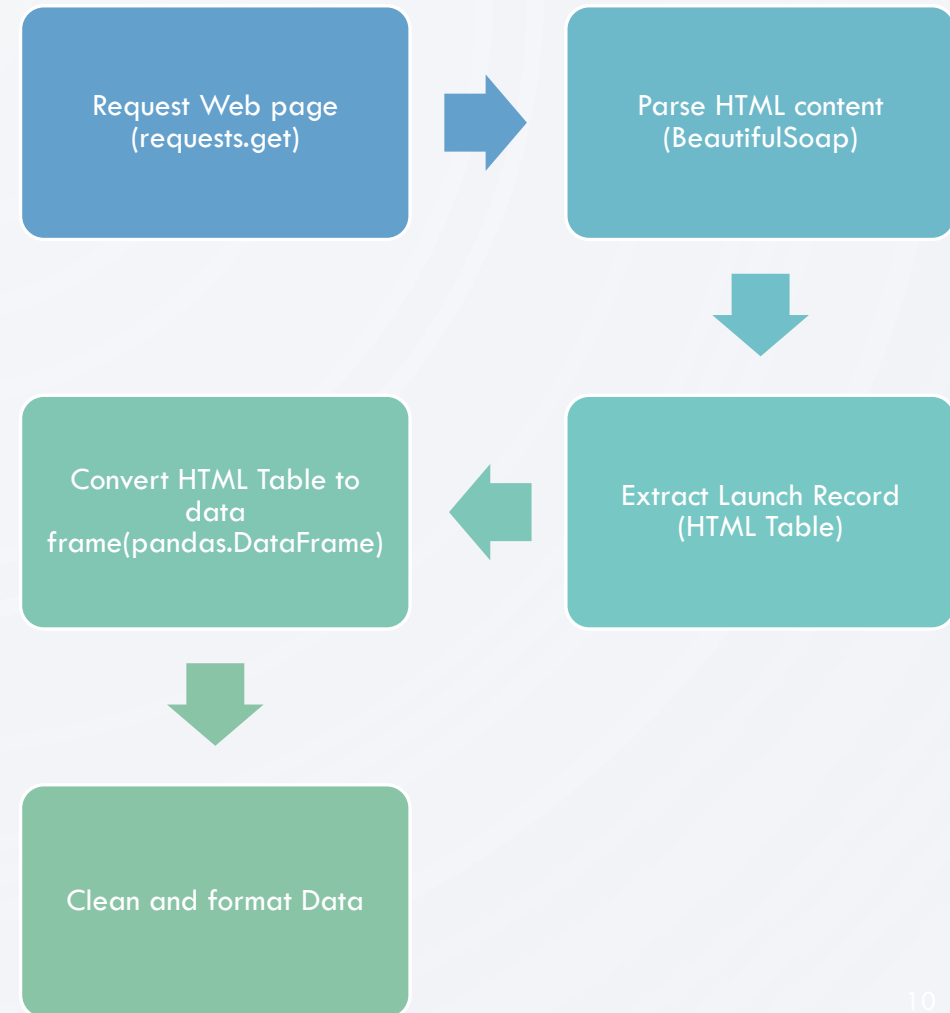
Step 3: Convert to Data Frame

Save the extracted HTML table into a pandas Data Frame.

Clean and format data frame for data consistency.

GitHub URL:

https://github.com/usamacaprian/IBM_Data_Science_Capstone/blob/main/jupyter-labs-webscraping.ipynb



Data Wrangling

Step 1: Data Cleaning

- Detect and handle missing values.
- Apply imputation methods or remove rows and columns with too many missing entries.

Step 2: Data Transformation

- Convert data types to correct formats such as date-time or numeric.
- Standardize text by using lowercase and trimming spaces.
- Create new features like extracting the year from the date column.
- Normalize or scale numerical data for consistency.

Data Wrangling

Step 3: Data Integration

- Combine data from multiple sources including API and web scraping.
- Align column names and data formats across datasets.

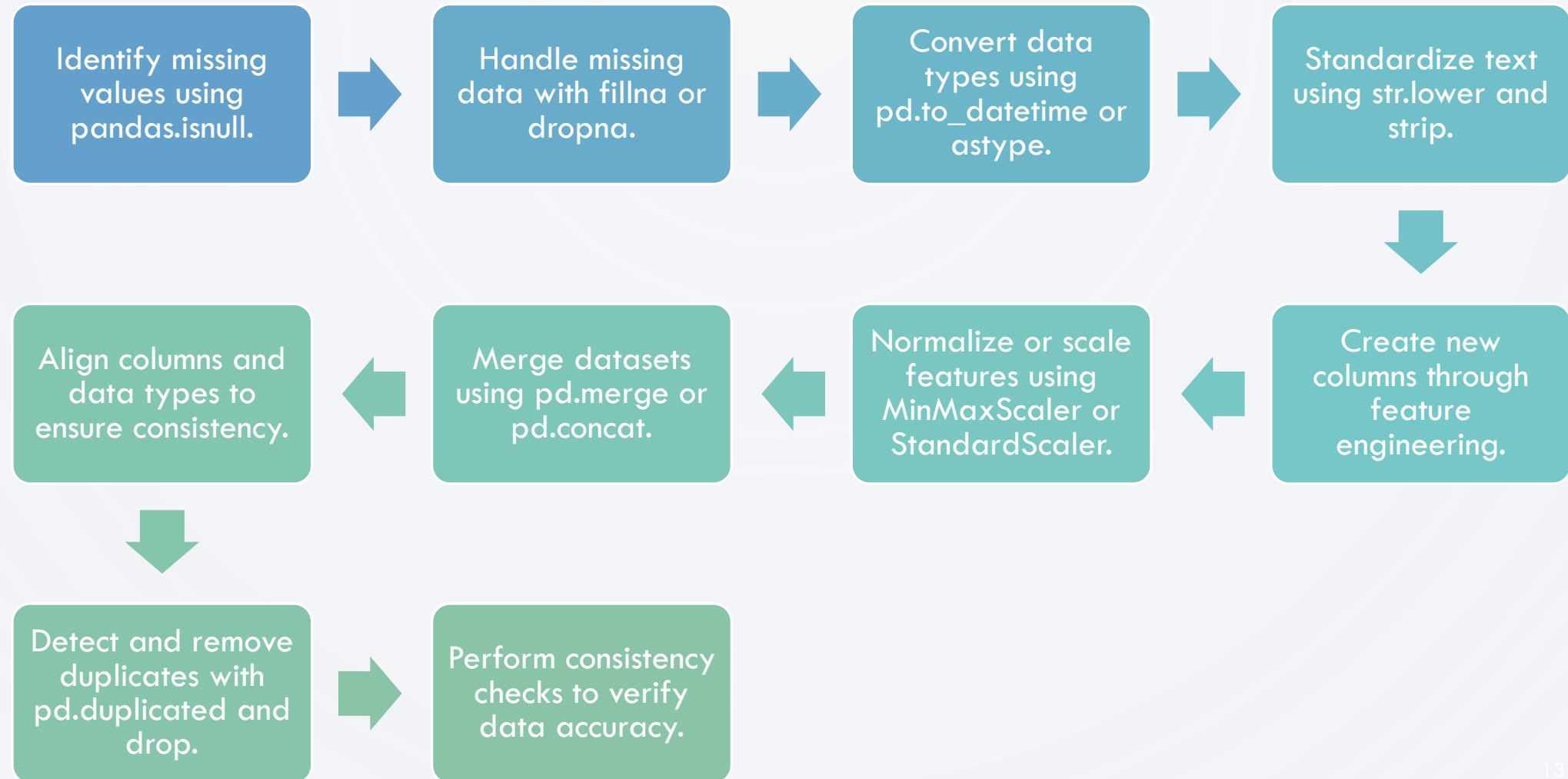
Step 4: Data Validation

- Identify and remove duplicates.
- Confirm data accuracy and consistency

GitHub URL:

[https://github.com/usamacaprian/IBM Data Science Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb](https://github.com/usamacaprian/IBM_Data_Science_Capstone/blob/main/labs-jupyter-spacex-Data%20wrangling.ipynb)

Data Wrangling – Flow Chart



EDA with Data Visualization

Overview

Exploratory Data Analysis (EDA) helps understand a dataset by summarizing and visualizing its main characteristics. It reveals distributions, patterns, and relationships between variables.

Charts Plotted

1. Histograms

- Show the distribution of numerical variables like payload mass and flight number.
- Help identify spread, outliers, and skewness.

2. Bar Charts

- Compare categorical variables such as launch outcomes across sites or rocket types.
- Highlight frequency patterns and category trends.

3. Line Charts

- Track changes over time, such as yearly launch success rates.
- Show performance trends and temporal shifts.

EDA with Data Visualization

4. Scatter Plots

- Display relationships between two numerical variables, like payload mass and success rate.
- Reveal correlations or dependencies between variables.

5. Heatmaps

- Visualize correlations among multiple numerical variables.
- Identify strong positive or negative relationships for feature selection.

6. Box Plots

- Show data distribution through quartiles.
- Highlight outliers and compare spreads across categories.

GitHub URL: [https://github.com/usamacaprian/IBM Data Science Capstone/blob/main/edadataviz.ipynb](https://github.com/usamacaprian/IBM_Data_Science_Capstone/blob/main/edadataviz.ipynb)

EDA with SQL

Aggregate Queries

- Calculated total launches.
- Counted successful and failed launches.
- Computed success rates by launch site and rocket type.

Join Queries

- Linked launch records with rocket details using table joins.
- Combined datasets for deeper analysis.

Filtering Queries

- Focused on specific launch outcomes such as success or failure.
- Applied filters based on launch date or rocket configuration.

EDA with SQL

Sorting Queries

- Sorted results to detect patterns or outliers.
- Ordered launches by date or success rate for better insight.

Subqueries

- Used nested queries to compute metrics like average payload mass per site.
- Enabled detailed analysis within broader datasets.
- **GitHub URL:** https://github.com/usamacaprian/IBM_Data_Science_Capstone/blob/main/jupyter-labs-eda-sql-coursera_sqlite.ipynb

Build an Interactive Map with Folium

Map Objects Created

Markers

- Added markers to show SpaceX launch sites on the map.
- Each marker represents a specific launch location.

Circles

- Placed circles around launch sites to indicate proximity zones.
- Used to visualize surrounding areas that may affect operations.

Lines

- Drew lines connecting launch sites and nearby locations.
- Provided spatial links between key points of interest.

Build an Interactive Map with Folium

Reasons for Adding Objects

Markers

- Identify exact launch locations for spatial context.
- Help users see where SpaceX conducts launches.

Circles

- Represent impact or safety zones around launch sites.
- Show operational boundaries visually.

Lines

- Display spatial relationships between sites and related features.
- Improve understanding of site connections and geography.

GitHub URL:

https://github.com/usamacaprian/IBM_Data_Science_Capstone/blob/main/lab_jupyter_launch_site_location.ipynb

Build a Dashboard with Plotly Dash

Plots and Graphs Added

Success Pie Chart

- Shows the share of successful and failed launches.
- Gives a clear view of overall mission performance.

Success-Payload Scatter Plot

- Displays the relationship between payload mass and launch success.
- Helps assess how payload weight impacts mission results.

Interactions Added

Launch Site Dropdown

- Lets users select and analyze specific launch sites.
- Focuses exploration by region or facility.

Range Slider for Payload

- Adjusts payload mass range interactively.
- Allows users to study how varying payloads affect success rates.

Build a Dashboard with Plotly Dash

Reasons for Adding Plots and Interactions

Success Pie Chart

- Offers a quick snapshot of mission success rates.
- Useful for evaluating overall performance metrics.

Success-Payload Scatter Plot

- Reveals trends between payload size and mission success.
- Supports better planning and operational analysis.

Launch Site Dropdown

- Improves user control and regional comparison.
- Enables location-based performance insights.

Range Slider for Payload

- Promotes detailed, dynamic payload analysis.
- Enhances understanding of payload effects on success.

GitHub URL: https://github.com/usamacaprian/IBM_Data_Science_Capstone/blob/main/spacex-dash-app.py

Predictive Analysis (Classification)

1. Data Preprocessing

- Standardized features so all variables contributed equally.
- Split the dataset into training and test sets for validation.

2. Model Selection

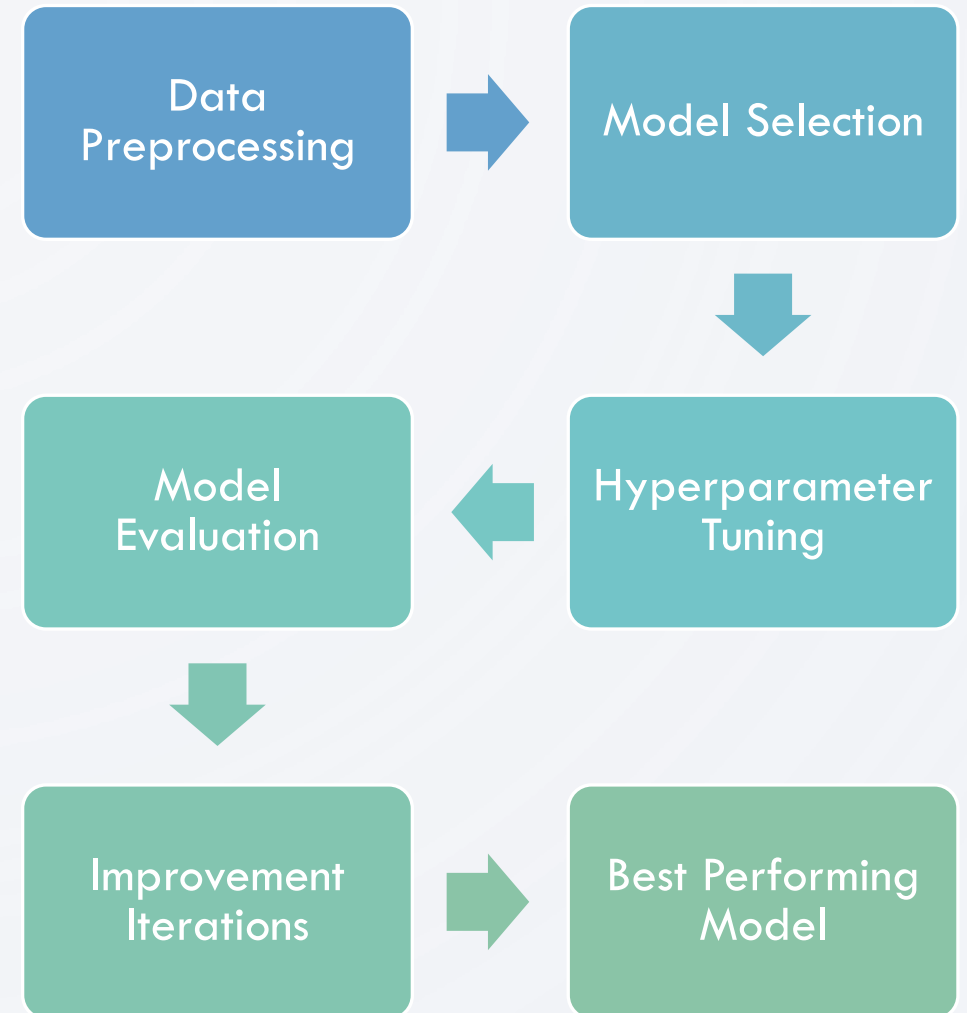
- Tested SVM, Decision Trees, and K-Nearest Neighbors (KNN).
- Selected models suited for binary classification based on project goals.

3. Hyperparameter Tuning

- Applied GridSearchCV to find the best hyperparameters.
- Tuned parameters like C (SVM), max_depth (Decision Trees), and n_neighbors (KNN).

4. Model Evaluation

- Used cross-validation to confirm model stability.
- Assessed performance using accuracy, precision, recall, and F1 -score.



Predictive Analysis (Classification)

5. Improvement Iterations

- Refined models using validation feedback.
- Adjusted parameters to boost prediction accuracy and consistency.

6. Best Model Selection

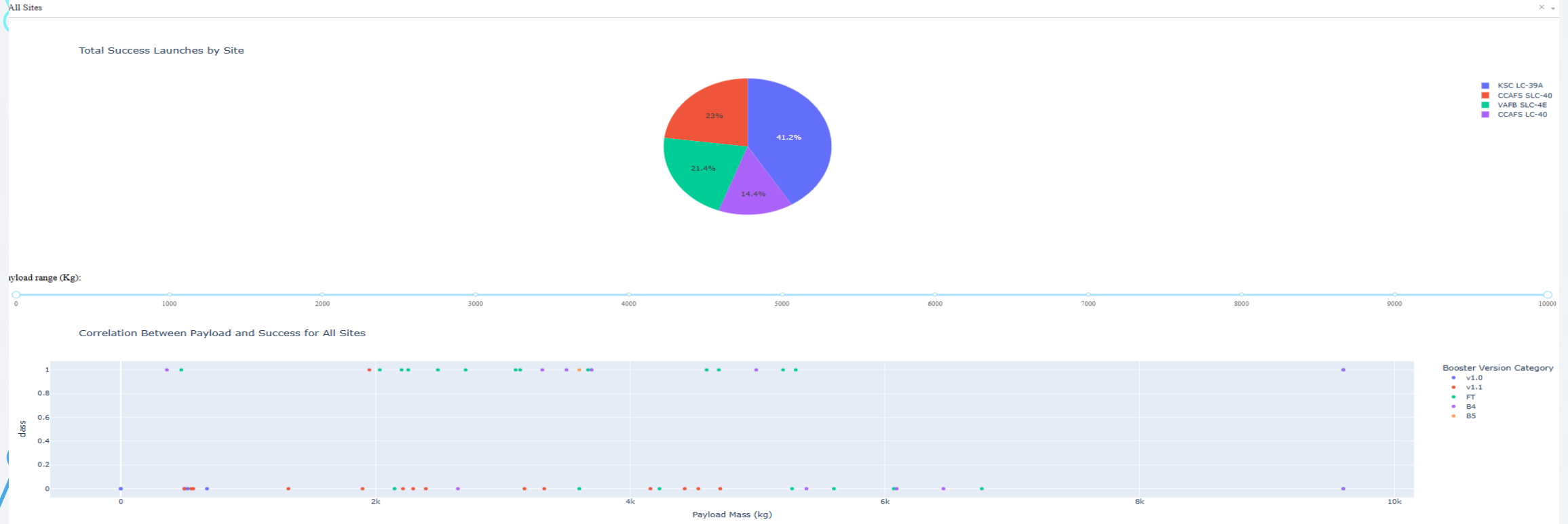
- Chose the model with the highest test accuracy.
- Ensured balanced performance between training and test results to prevent overfitting.

GitHub URL:

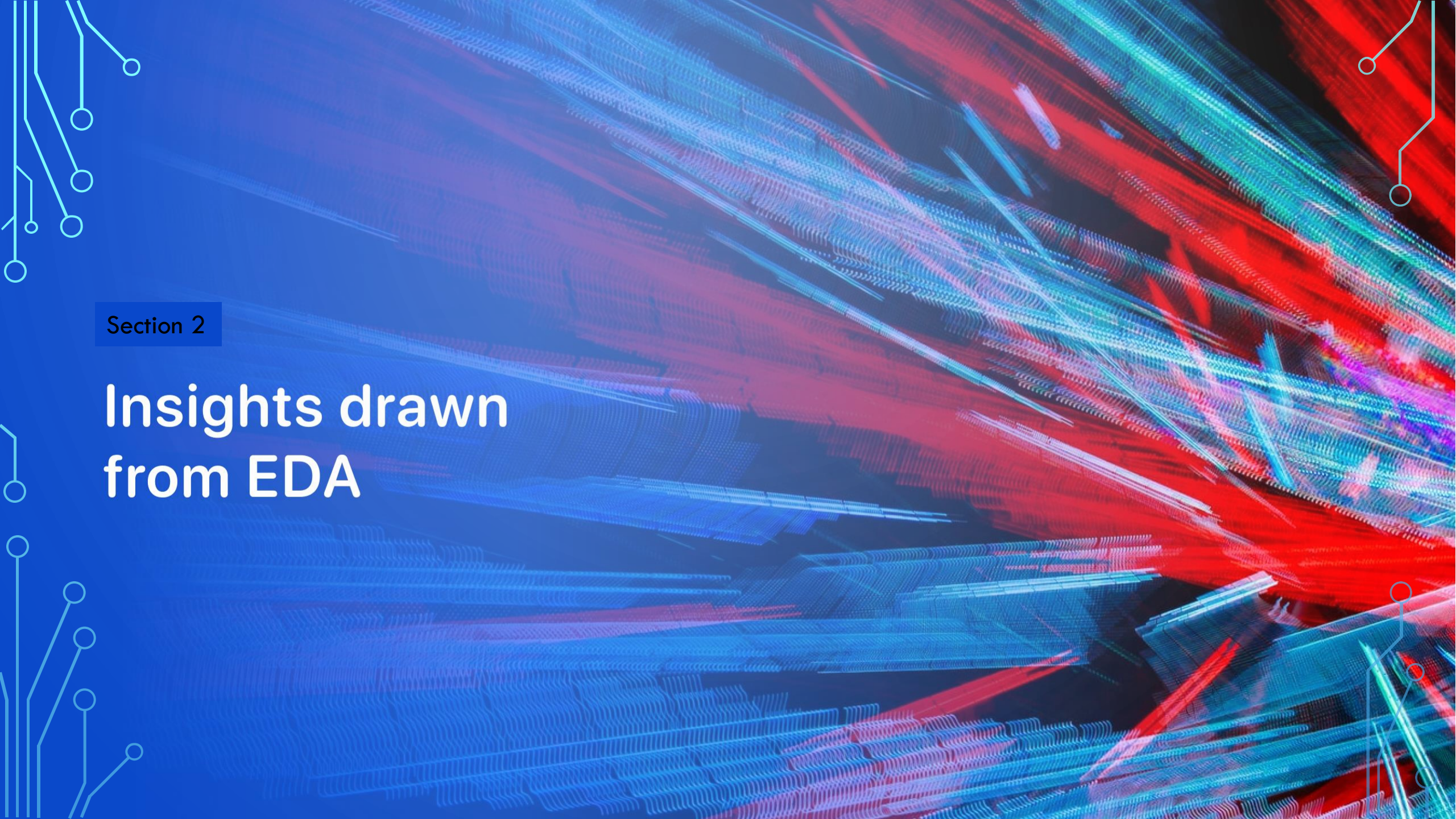
https://github.com/usamacaprian/IBM_Data_Science_Capstone/blob/main/SpaceX_Machine%20Learning%20Prediction_Part_5.ipynb

Results

SpaceX Launch Records Dashboard



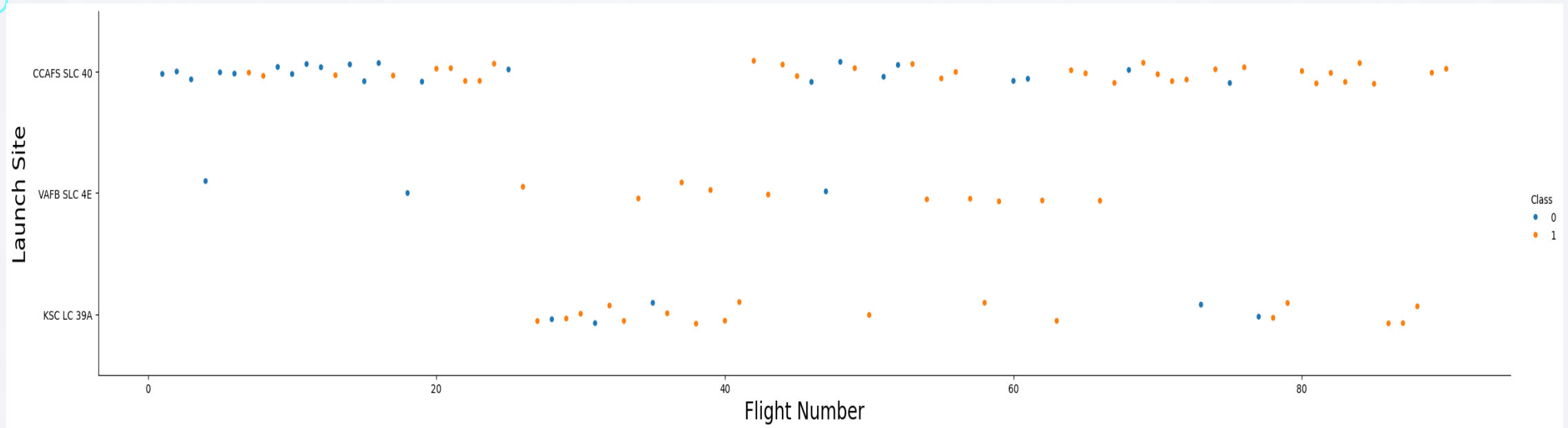
This is a preview of the Plotly dashboard. The next slides present results from EDA with visualization, EDA with SQL, the interactive map built using Folium, and the final model outcomes.



Section 2

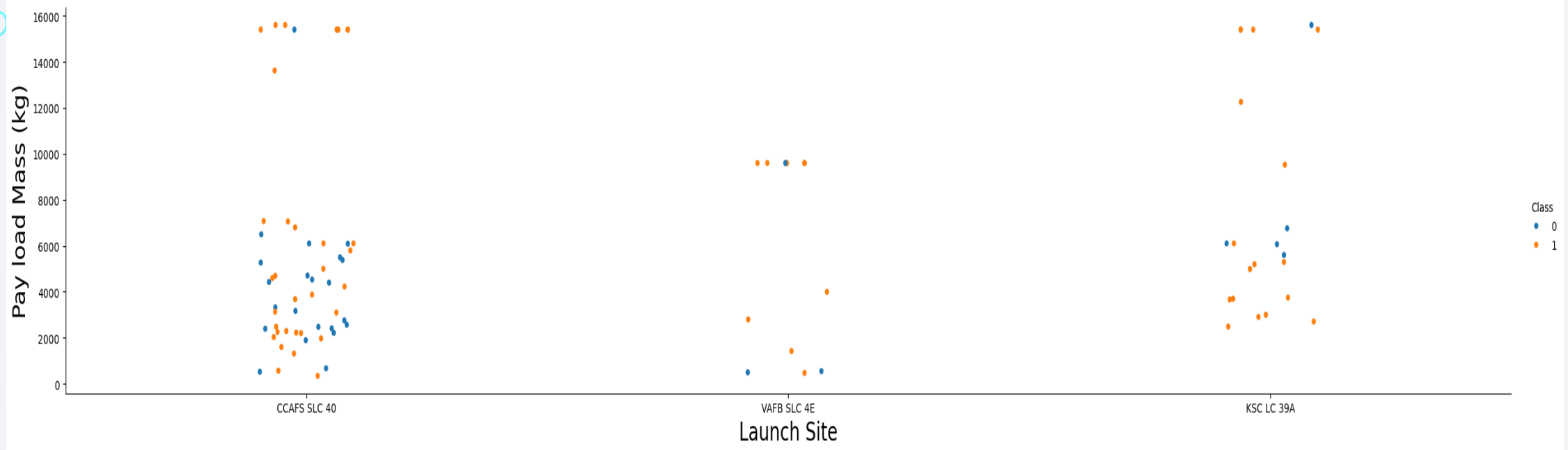
Insights drawn from EDA

Flight Number vs. Launch Site



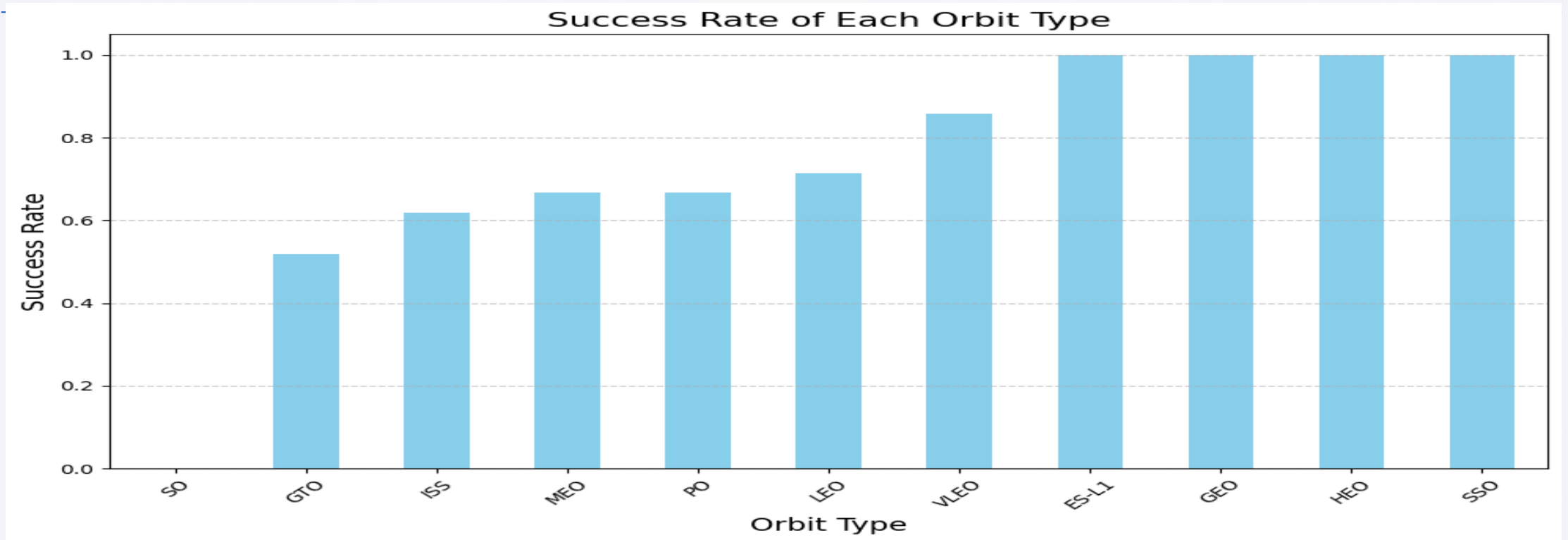
- Mixed outcomes at key launch sites, including CCAFS SLC 40 and KSC LC 39A, show both successful (orange) and failed (blue) landings, suggesting landing success depends on factors beyond the launch site.
- Launches occur across a wide range of flight numbers at all sites, showing steady activity over time without a clear improvement or decline in landing success.

Payload vs. Launch Site



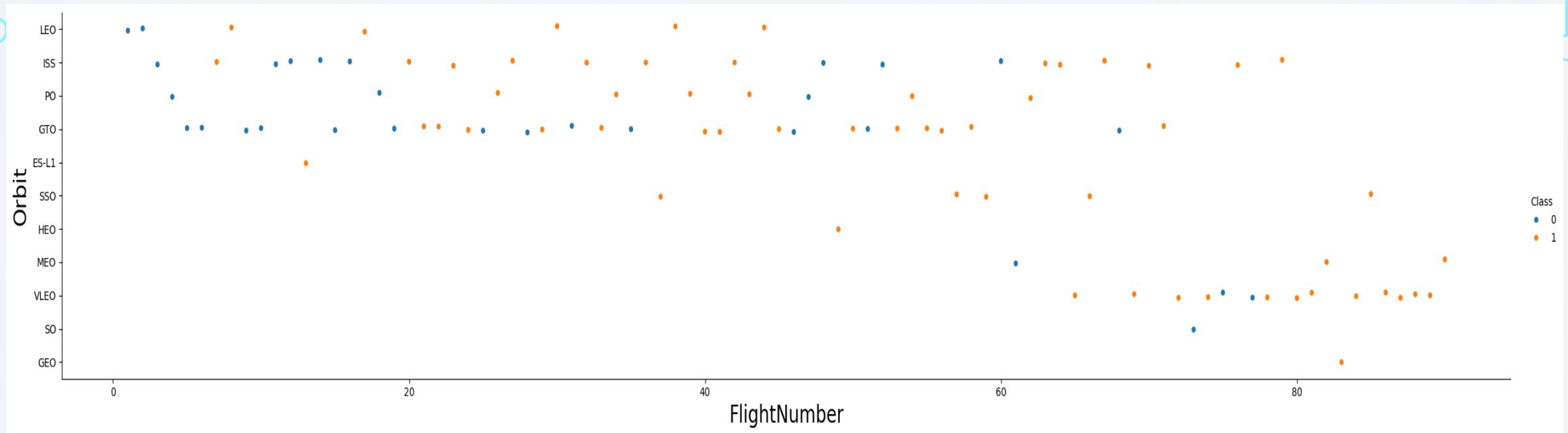
- Most launches from CCAFS SLC 40 carry payloads under 10,000 kg, while VAFB SLC 4E and KSC LC 39A handle a broader payload range, showing diverse mission types.
- KSC LC 39A often supports heavier payloads above 15,000 kg, highlighting its role in high-capacity missions.

Success Rate vs. Orbit Type



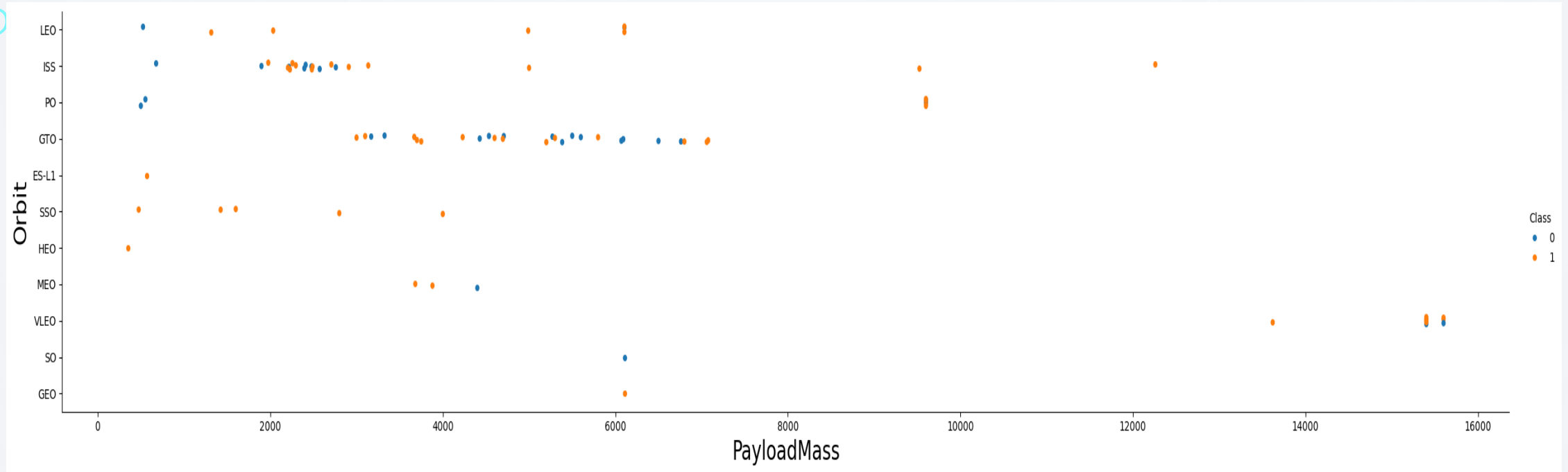
- Missions to VLEO, ES-L1, GEO, HEO, and SSO orbits show perfect first-stage landing success, indicating high reliability for these orbit types.
- GTO missions have a notably lower success rate, reflecting higher difficulty and operational challenges for this orbit.

Flight Number vs. Orbit Type



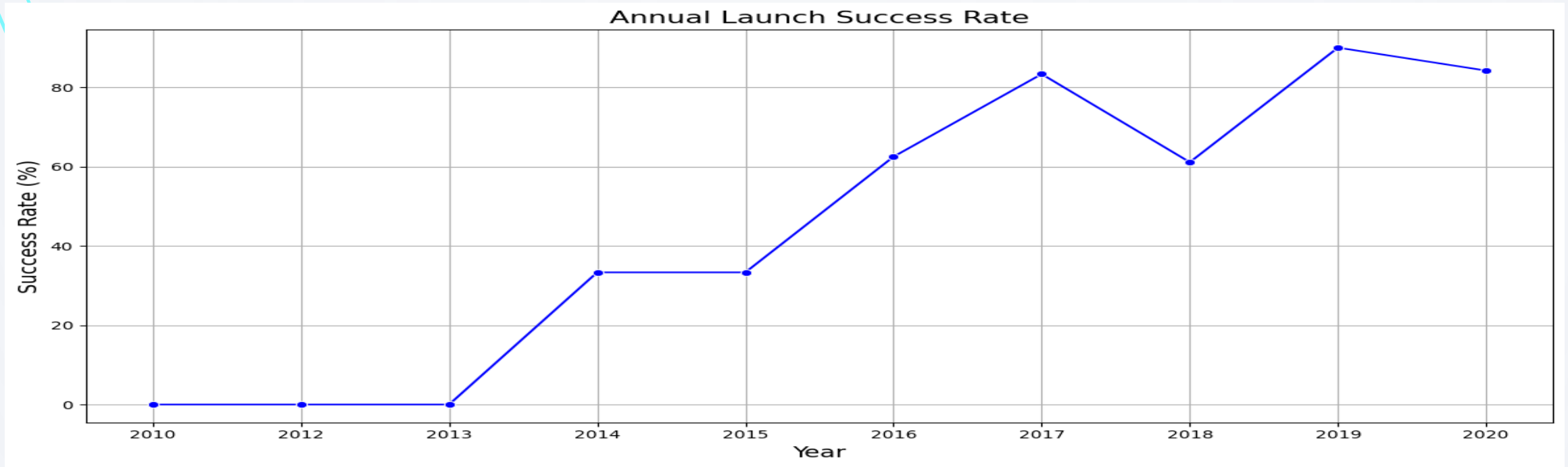
- Falcon 9 launch success rates increase with higher flight numbers, showing that experience and design improvements lead to better results.
- Earlier GTO and ISS missions had mixed outcomes, while recent ones show higher success, indicating progress in mission planning and execution

Payload vs. Orbit Type



- Successful landings are more common across all orbit types for payloads under 6000 kg.
- Payloads above 10,000 kg show mixed outcomes, suggesting higher difficulty with heavier launches.

Launch Success Yearly Trend



- Annual launch success rates have risen sharply since 2013, surpassing 80% by 2020.
- Although there was a decline in 2018, the overall trend shows steady improvement in Falcon 9 reliability and performance.

All Launch Site Names

Task 1

Display the names of the unique launch sites in the space mission

```
[16]: %sql select DISTINCT LAUNCH_SITE from SPACEXTABLE
* sqlite:///my_data1.db
Done.
```

```
[16]: Launch_Site
-----
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40
```

- Displaying the names of the unique launch sites in the space mission.

Launch Site Names Begin with 'CCA'

Display 5 records where launch sites begin with the string 'CCA'

```
[17]: %sql SELECT * FROM SPACEXTABLE WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

[17]:

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

- Displaying 5 records where launch sites begin with the string 'CCA'

Total Payload Mass

Display the total payload mass carried by boosters launched by NASA (CRS)

```
[18]: %sql SELECT SUM("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Customer" = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

Done.

```
[18]: SUM("PAYLOAD_MASS_KG_")  
-----  
45596
```

- Displaying the total payload mass carried by boosters launched by NASA (CRS).

Average Payload Mass by F9 v1.1

Display average payload mass carried by booster version F9 v1.1

```
[19]: %sql SELECT AVG("PAYLOAD_MASS_KG_") FROM SPACEXTABLE WHERE "Booster_Version" = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

Done.

```
[19]: AVG("PAYLOAD_MASS_KG_")
```

2928.4

- Displaying average payload mass carried by booster version F9 v1.1.

First Successful Ground Landing Date

List the date when the first succesful landing outcome in ground pad was acheived.

Hint: Use min function

```
0]: %sql SELECT MIN("Date") FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (ground pad)';
```

```
* sqlite:///my_data1.db
```

Done.

```
0]: MIN("Date")
```

```
2015-12-22
```

- Listing the date when the first successful landing outcome in ground pad was achieved.

Successful Drone Ship Landing with Payload between 4000 and 6000

List the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000

```
[21]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "Landing_Outcome" = 'Success (drone ship)' AND "PAYLOAD_MASS_KG_" > 4000 AND "PAYLOAD_MASS_KG_" < 6000;
```

```
* sqlite:///my_data1.db
```

Done.

```
[21]: Booster_Version
```

```
F9 FT B1022
```

```
F9 FT B1026
```

```
F9 FT B1021.2
```

```
F9 FT B1031.2
```

- Listing the names of the boosters which have success in drone ship and have payload mass greater than 4000 but less than 6000.

Total Number of Successful and Failure Mission Outcomes

List the total number of successful and failure mission outcomes

```
[22]: %sql SELECT "Mission_Outcome", COUNT(*) AS "Total" FROM SPACEXTABLE WHERE "Mission_Outcome" IN ('Success', 'Failure') GROUP BY "Mission_Outcome";
```

```
* sqlite:///my_data1.db
```

Done.

```
[22]: Mission_Outcome  Total
```

Mission_Outcome	Total
Success	98

- Listing the total number of successful and failure mission outcomes.

Boosters Carried Maximum Payload

List all the booster_versions that have carried the maximum payload mass, using a subquery with a suitable aggregate function.

```
[23]: %sql SELECT DISTINCT "Booster_Version" FROM SPACEXTABLE WHERE "PAYLOAD_MASS_KG_" = (SELECT MAX("PAYLOAD_MASS_KG_") FROM SPACEXTABLE);  
* sqlite:///my_data1.db  
Done.
```

```
[23]: Booster_Version
```

F9 B5 B1048.4

F9 B5 B1049.4

F9 B5 B1051.3

F9 B5 B1056.4

F9 B5 B1048.5

F9 B5 B1051.4

F9 B5 B1049.5

F9 B5 B1060.2

F9 B5 B1058.3

F9 B5 B1051.6

F9 B5 B1060.3

F9 B5 B1049.7

- Listing the names of the booster versions which have carried the maximum payload mass.

2015 Launch Records

```
SELECT
CASE
  WHEN substr("Date", 6, 2) = '01' THEN 'January'
  WHEN substr("Date", 6, 2) = '02' THEN 'February'
  WHEN substr("Date", 6, 2) = '03' THEN 'March'
  WHEN substr("Date", 6, 2) = '04' THEN 'April'
  WHEN substr("Date", 6, 2) = '05' THEN 'May'
  WHEN substr("Date", 6, 2) = '06' THEN 'June'
  WHEN substr("Date", 6, 2) = '07' THEN 'July'
  WHEN substr("Date", 6, 2) = '08' THEN 'August'
  WHEN substr("Date", 6, 2) = '09' THEN 'September'
  WHEN substr("Date", 6, 2) = '10' THEN 'October'
  WHEN substr("Date", 6, 2) = '11' THEN 'November'
  WHEN substr("Date", 6, 2) = '12' THEN 'December'
  ELSE 'Unknown'
END AS "Month_Name",
"Mission_Outcome",
"Booster_Version",
"Launch_Site"
FROM
SPACEXTABLE
WHERE
substr("Date", 0, 5) = '2015';

* sqlite:///my_data1.db
Done.
```

[28] :

Month_Name	Mission_Outcome	Booster_Version	Launch_Site
January	Success	F9 v1.1 B1012	CCAFS LC-40
February	Success	F9 v1.1 B1013	CCAFS LC-40
March	Success	F9 v1.1 B1014	CCAFS LC-40
April	Success	F9 v1.1 B1015	CCAFS LC-40
April	Success	F9 v1.1 B1016	CCAFS LC-40
June	Failure (in flight)	F9 v1.1 B1018	CCAFS LC-40
December	Success	F9 FT B1019	CCAFS LC-40

- Listing the failed landing outcomes in drone ship, their booster versions and launch site names for the months in year 2015.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order.

```
[27]: %%sql
SELECT
    "Landing_Outcome",
    COUNT(*) AS "Count"
FROM
    SPACEXTABLE
WHERE
    "Date" BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY
    "Landing_Outcome"
ORDER BY
    COUNT(*) DESC;
```

```
* sqlite:///my_data1.db
Done.
```

```
[27]:
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

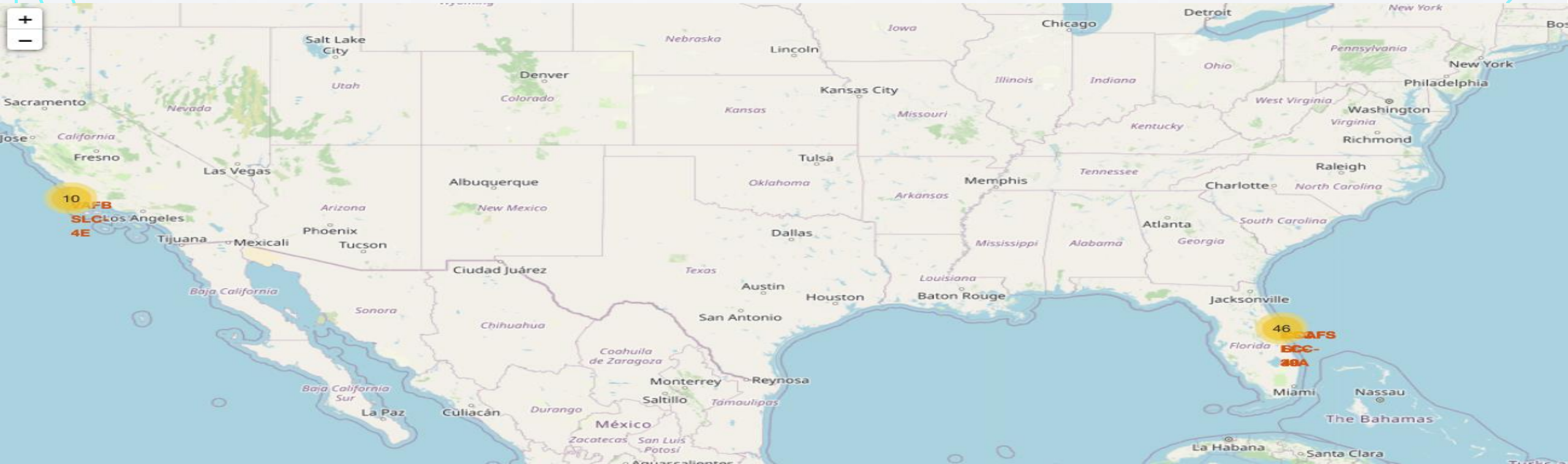
- Ranking the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20 in descending order.



Section 3

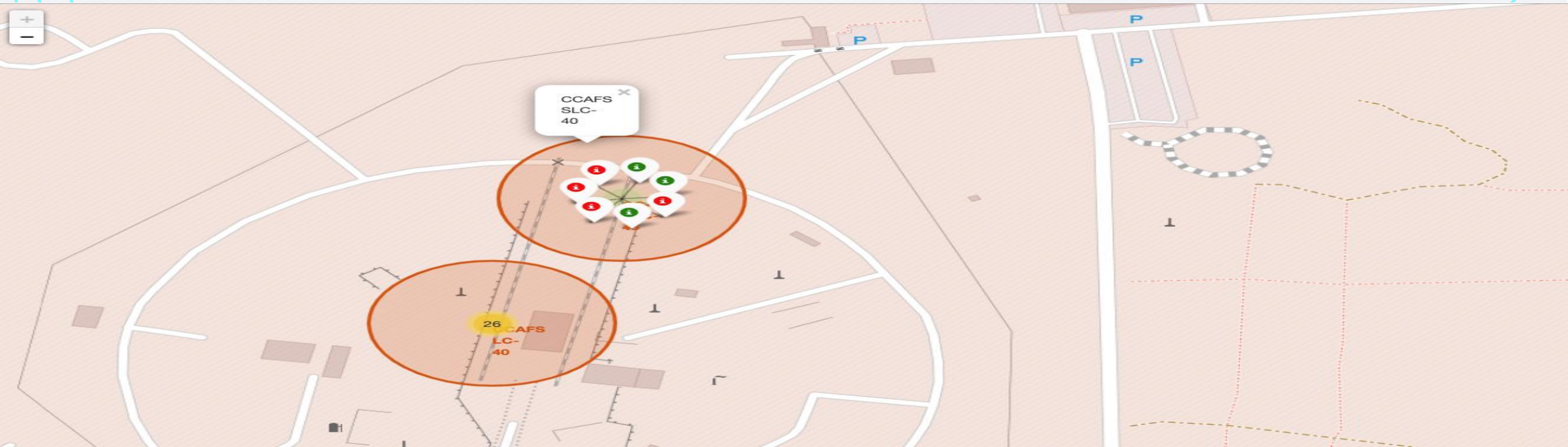
Launch Sites Proximities Analysis

All launch sites' location markers on a global map



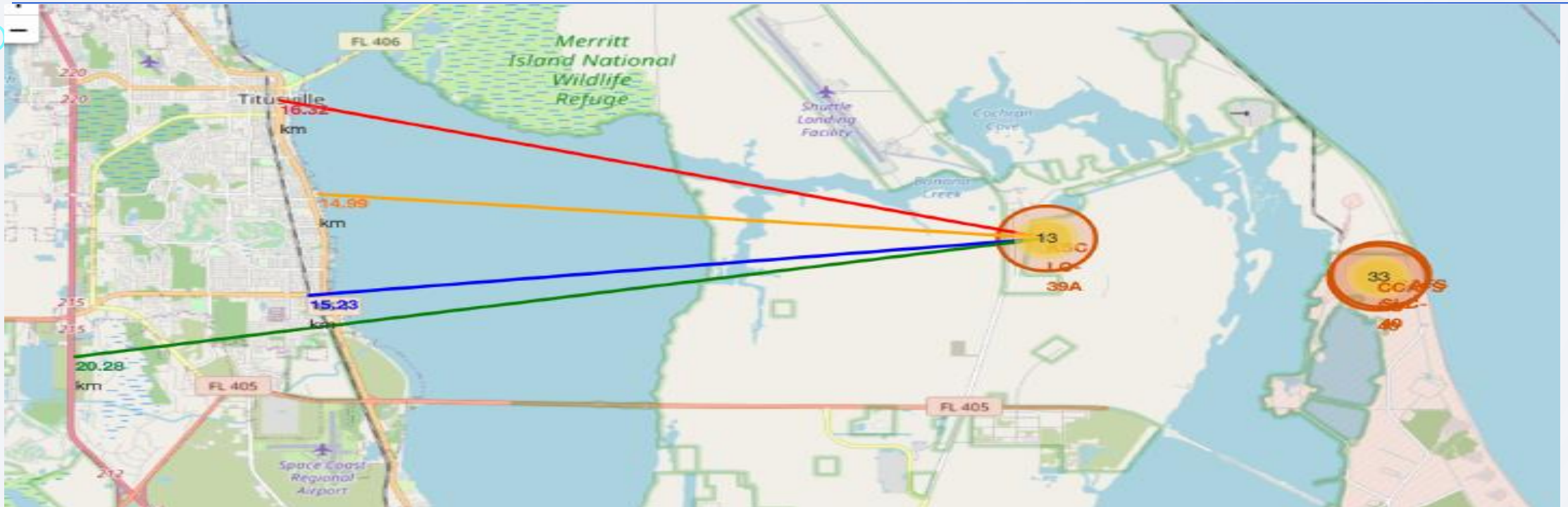
- Most launch sites are located near the equator, where the Earth's rotation speed is about 1670 km/h. This rotational speed gives rockets an extra boost due to inertia, helping them reach orbit more efficiently.
- Launch sites are positioned close to coastlines so rockets can launch over the ocean, reducing the risk of debris falling near populated areas.

Colour-labeled launch records on the map

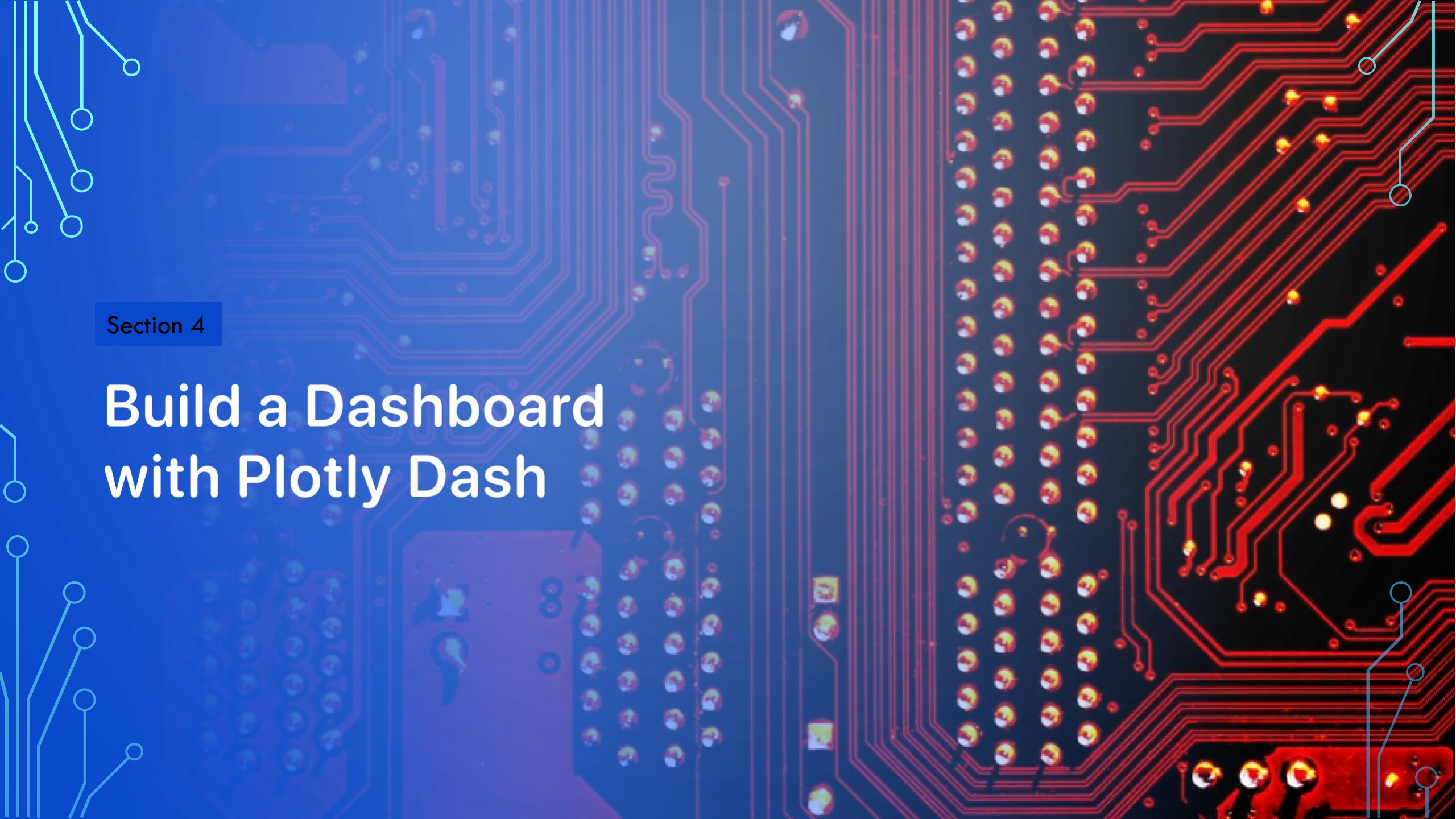


- This visualization with clustered markers makes SpaceX launch data easier to study. Clustering helps manage overlapping points and reveals patterns across sites. The marker colors and popups show detailed information about each launch.
- For instance, in the screenshot for CCAFS LC-40, there are 26 launches with 19 red and 7 green markers. Red markers show failed launches, while green show successful ones. This quick color view helps assess performance trends for each launch site.

Distances between a launch site to its proximities



- This plot shows how close the CCAFS SLC-40 launch site is to the coastline. The straight-line distance is about 0.51 kilometers, marked by the PolyLine. The short distance reflects standard launch site placement near coasts. This allows rockets to follow over-water paths and reduces risk to nearby populations during launches or debris fall.



Section 4

Build a Dashboard with Plotly Dash

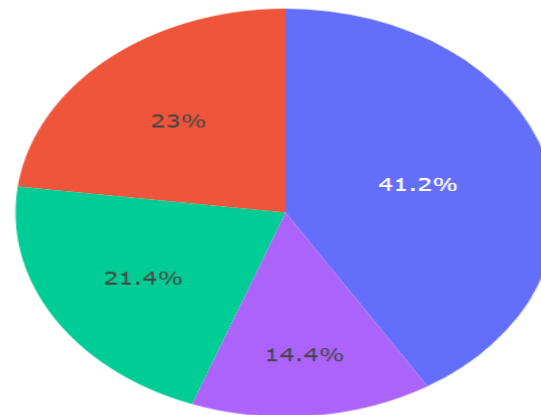
Launch success count for all sites

SpaceX Launch Records Dashboard

All Sites



Total Success Launches by Site



■ KSC LC-39A
■ CCAFS SLC-40
■ VAFB SLC-4E
■ CCAFS LC-40

- The chart clearly shows that from all the sites, KSC LC-39A has the most successful launches.

Launch site with highest launch success ratio

Total Success Launches for Site KSC LC-39A



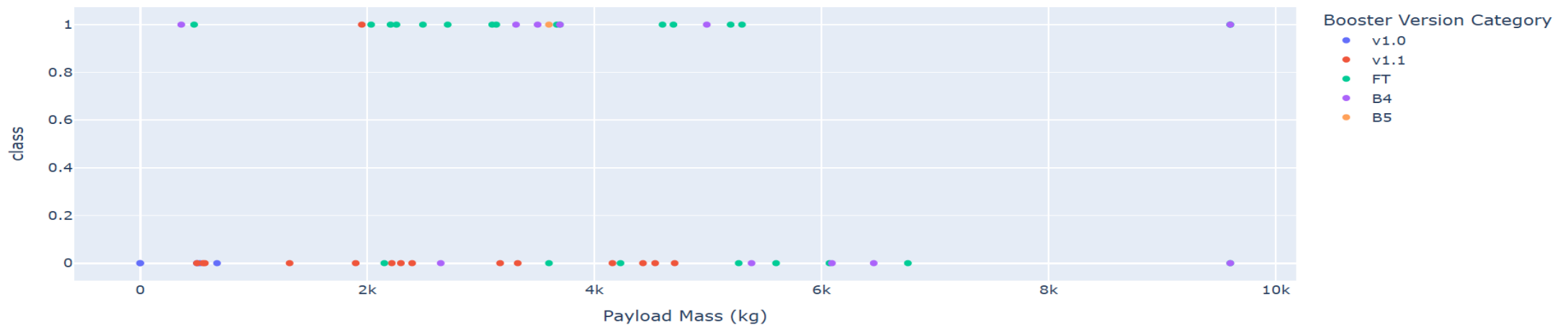
- KSC LC-39A has the highest launch success rate (76.9%) with 10 successful and only 3 failed landings.

Payload Mass vs. Launch Outcome for all sites

Payload range (Kg):



Correlation Between Payload and Success for All Sites



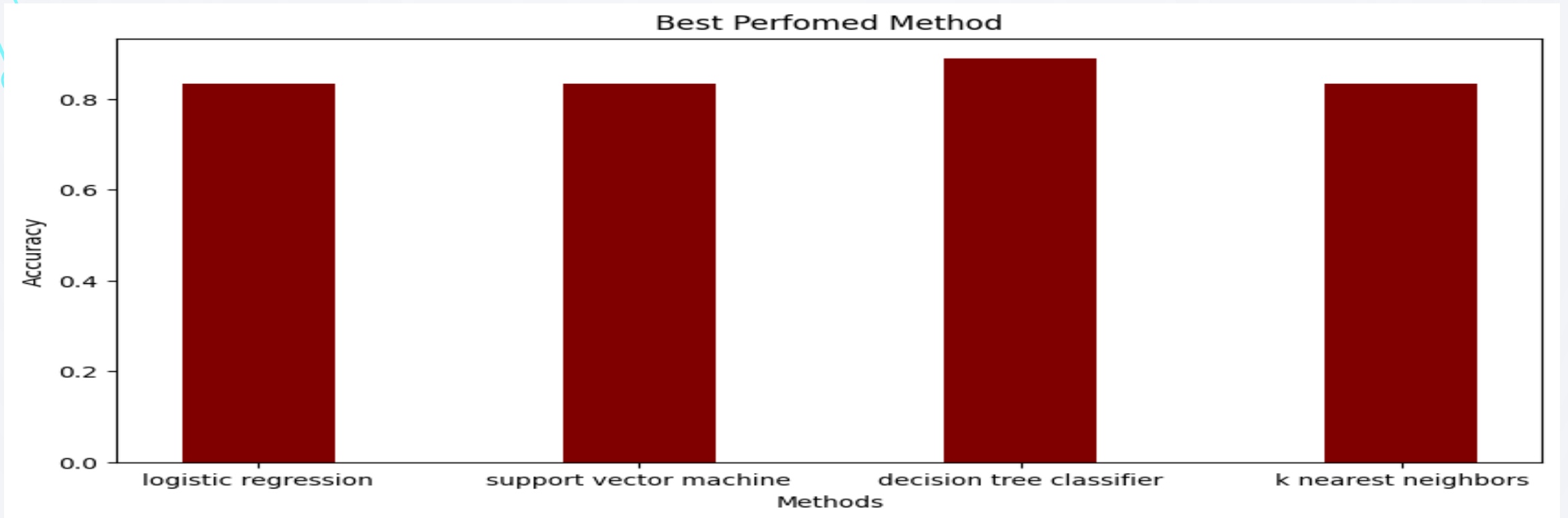
- The charts show that payloads between 2000 and 5500 kg have the highest success rate.



Section 5

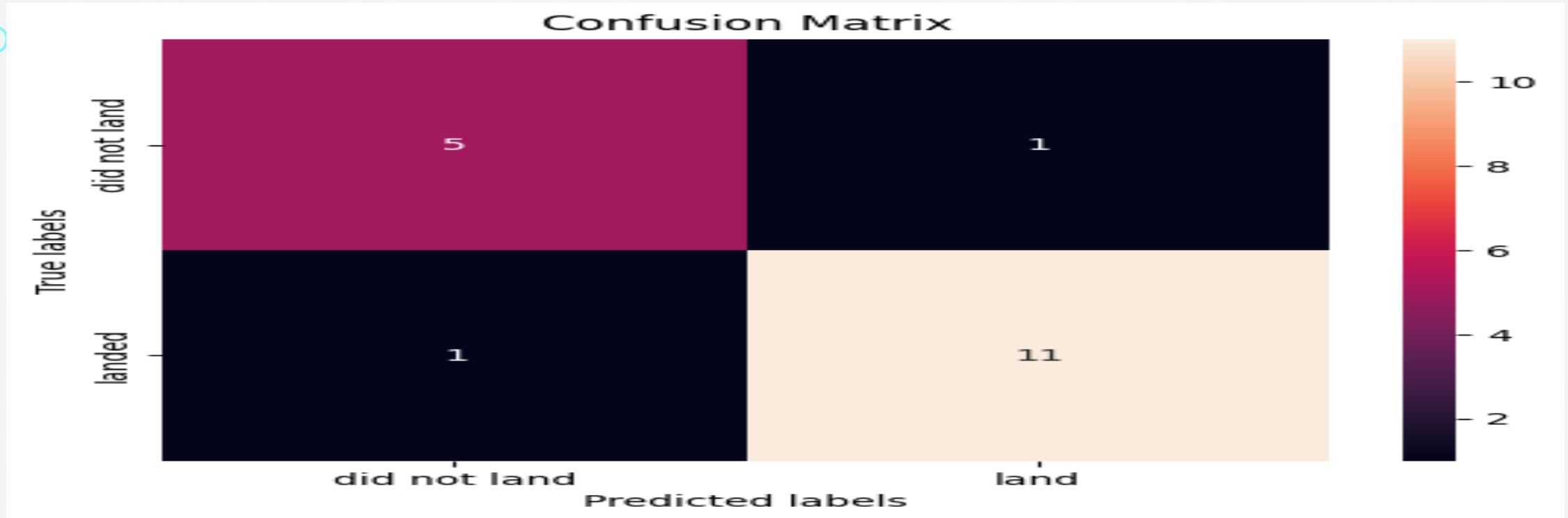
Predictive Analysis (Classification)

Classification Accuracy



- The Decision Tree model achieved the highest test accuracy of 0.888. This indicates it fits the dataset better than Logistic Regression, Support Vector Machine, and K Nearest Neighbors, which each reached 0.8333 accuracy.

Confusion Matrix



- The confusion matrix shows that logistic regression can separate the classes, but it produces many false positives. This reduces its reliability despite overall reasonable classification performance.

Conclusions

- The Decision Tree model gives the best accuracy among all tested algorithms.
- Launches with lower payload mass perform better than those with heavier payloads.
- Most launch sites are near the Equator, gaining a velocity advantage from Earth's rotation.
- All launch sites are located close to coastlines, reducing risk to populated areas.
- Launch success rates have improved steadily over the years.
- KSC LC-39A records the highest success rate among all launch sites.
- Orbits ES-L1, GEO, HEO, and SSO show a 100% success rate.

Appendix

- GitHub repository url:
 - https://github.com/usamacaprian/IBM_Data_Science_Capstone
- Special Thanks to All Instructors:
 - <https://www.coursera.org/professional-certificates/ibm-data-science?#instructors>

Thank you!

